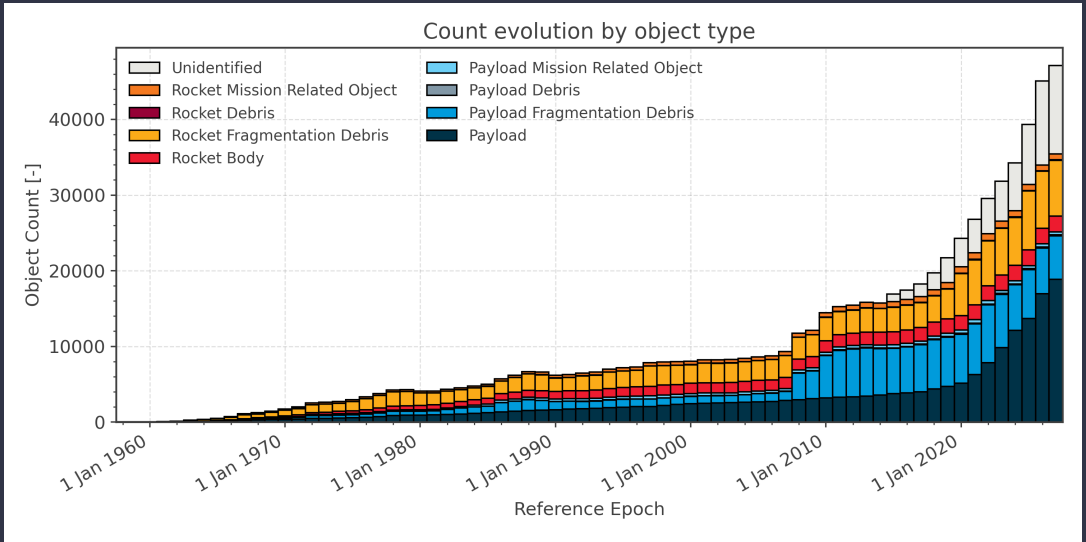
Deep learning-based classification of resident space objects with non-periodic light curves

Zhao et al. (2026)

An analysis and overview of the state of the field by me :^)

Introduction

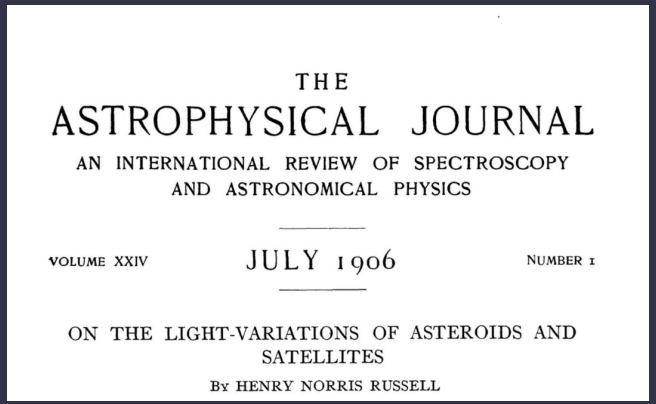
- Resident Space Objects (RSOs)
  - Satellites
  - Rocket Bodies
  - Debris
  - Literally anything!
- Space Situational Awareness
  - Objects too far to image
- Cataloging vs Categorization
  - Ephemeris (Orbit)
  - Attitude
  - Shape



Number of RSOs over time [2]

Background

- Light-curve: time-series of reflected brightness
- Mathematical models to determine attitude and shape since 1906 [3]
- "If certain characteristics of an RSO are known, such as its shape and surface reflectance properties, other target attributes can be inferred through light-curve inversion" [1]
  - Properties can still be inferred without pre-existing knowledge
- Modern methodologies:
  - Filtering
  - Machine learning
    - Recurrent neural networks (RNN)
    - Long short term memory (LSTM)
    - One-dimensional convolutional neural network (1D-CNN)
  - Frequency spectrum analysis



THE
ASTROPHYSICAL JOURNAL
AN INTERNATIONAL REVIEW OF SPECTROSCOPY AND ASTRONOMICAL PHYSICS

VOLUME XXIV JULY 1906 NUMBER 1

ON THE LIGHT-VARIATIONS OF ASTEROIDS AND SATELLITES

By HENRY NORRIS RUSSELL

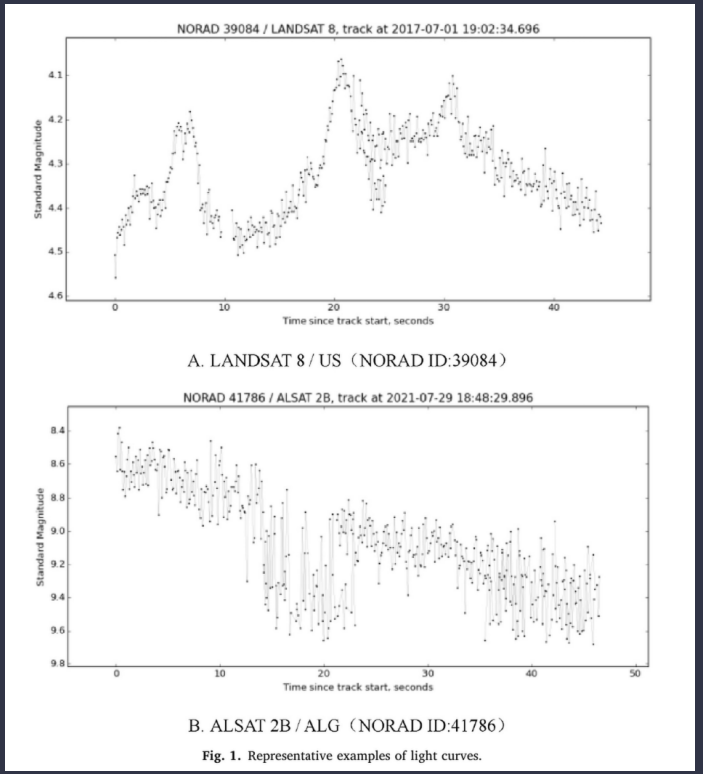


Fig. 1. Representative examples of light curves.

Previous work

- Lu et al
  - Selected periodic satellite and rocket bodies from MMT
  - Interpolated with filters to achieve uniform sampling
  - CNN (binary classification)
- Liu et al
  - Zero-padding for missing data
  - LSTM-NN (four classes)
- Linares et al
  - Zero-padding for missing data
  - CNN (3 classes)
- Simon et al
  - Least Squares
  - Empirical mode decomposition
  - CoAtNet (3 classes)
- Most studies have been on periodic data
- Most objects aren't periodic (>1% in MMT) [1]
- Assumption of uniform time intervals and fixed input size

Data

- Mini-Mega-TORTORA (MMT)
  - Measured from wide-field ground systems
  - Provides:
    - Light curves in apparent and standardized magnitudes
    - Time-varying distance and phase angle
    - If satellite was in penumbra (partial shadow or eclipse)
    - Period if periodic
  - Missing values
- RoBo6
  - Processed from MMT using Light Curve Data Creator (LCDC)
  - 6 classes of bodies
- Space Debris Light Curve Database (SDLCD)
  - Focuses on objects in GEO, eccentric (GTO, Molniya) and GNSS (aka GPS aka MEO)
- MMT has the most data, and has the least processing
- Want to avoid synthetic generated data
- Many light curves are unique
- It is difficult to monitor the same satellite consistently
- Sampled data from MMT is cut by track
  - Irregular sampling
  - Variable Length
- Out of all the parameters, 6 are used for features:
  - Magnitude
  - STD magnitude
  - Filter
  - Penumbra
  - Distance
  - Phase



** Mini-MegaTORTORA imaging system [4]**

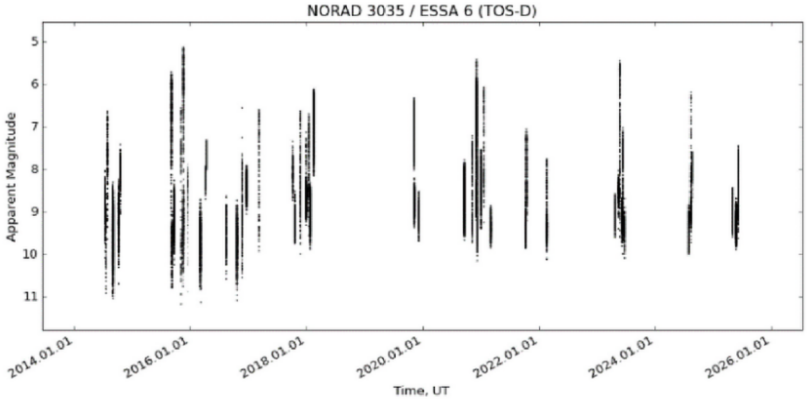


Fig. 2. Complete light curve data for ESSA 6 (NORAD ID: 3035).

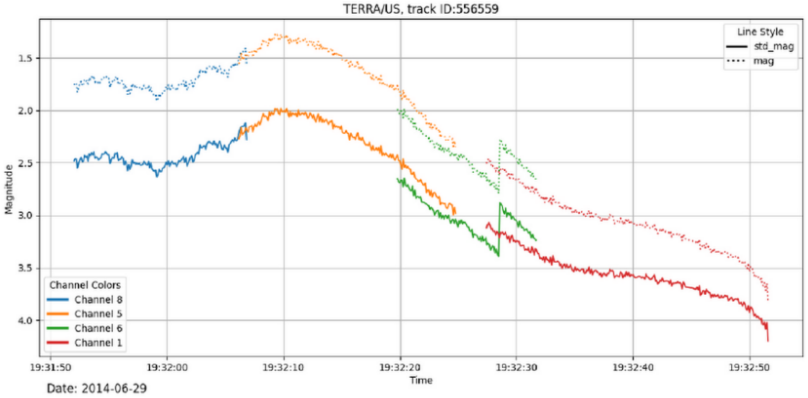


Fig. 3. The light curve of the U.S. TERRA satellite (NORAD ID: 25994) for track 556559.

** Figure 2 shows multiple tracks, and Figure 3 shows the multiple channels that make up a track [1] **

Models

- Biggest sources of irregularly sampled and multivariate data are from healthcare, economics, and earth sciences [1]
- These two were selected based on their performance on medical field data
- Time-Parameterized Convolution Neural Network (TPCNN)
- Multi-Time Attention Module Discretization (mTAND) Network

Time-Parameterized Convolutional Neural Network

- Standard CNNs assume time series is uniform
  - Kernel slides over inputs with fixed weights
- Time is added in the embedding construction
- In the first CONV1D block weights are adjusted based on time delta
- Output of encoder passed into a standard NN

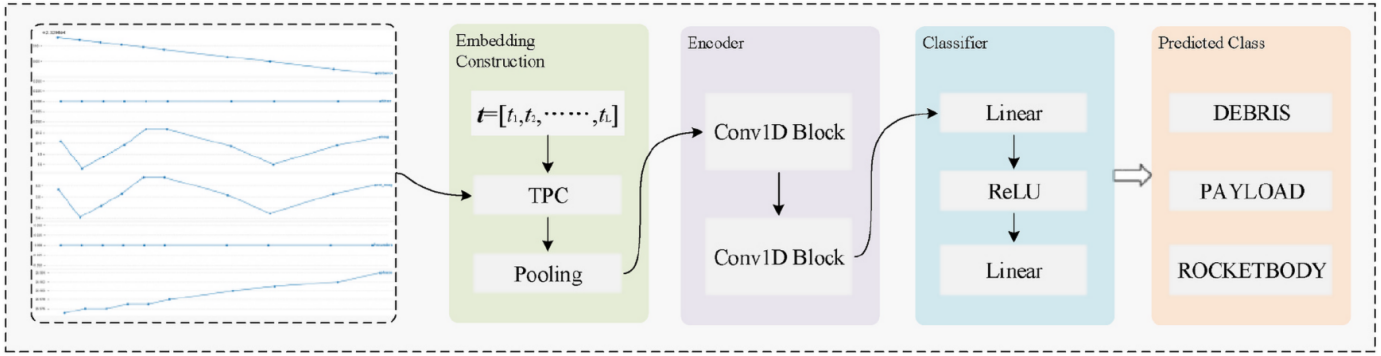


Fig. 7. Schematic diagram of the TPCNN architecture.

Multi-Time Attention Module Discretization Network

- r vector are reference time points
- mTAND module takes irregularly sampled time series and outputs continuous time embeddings based on r
- Attention score is determined based off of time difference between current r and all light curve time values
- Attention weight is generated at runtime
- Unlike normal attention based NN, classification output means decoder is dropped
- GRU is a RNN that takes in the sequential output of the embedding construction and outputs a final hidden state

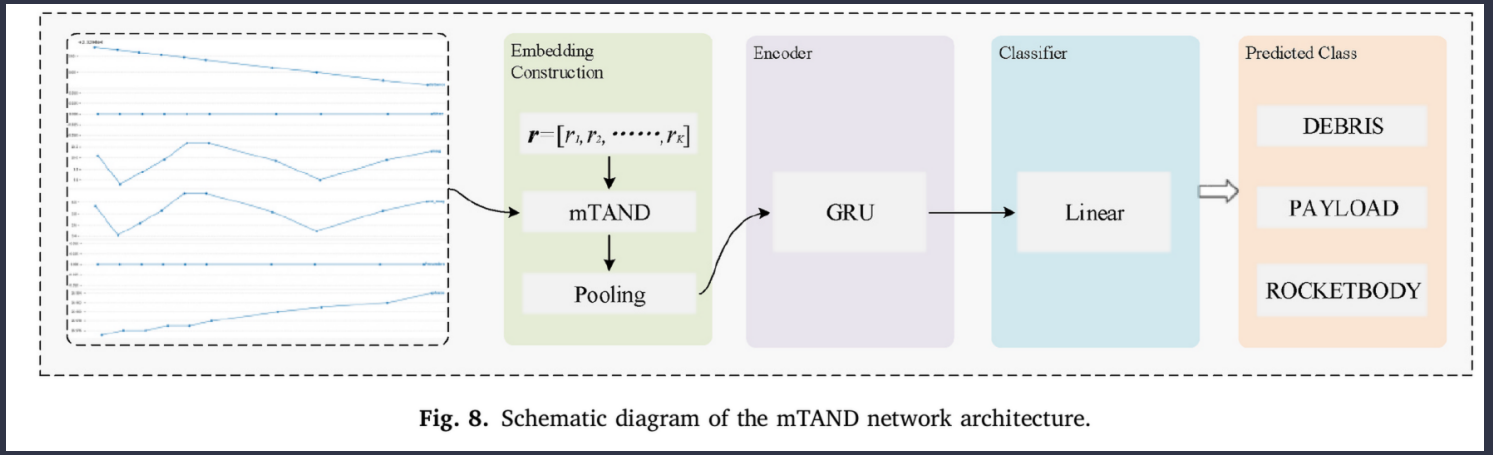


Fig. 8. Schematic diagram of the mTAND network architecture.

Improved mTAND

- Feature-wise gated attention module
- Not all features are equally important at every time step
  - Use sigmoid to modify them based on time
- Dropout blocks prevent overfitting by dropping values during training

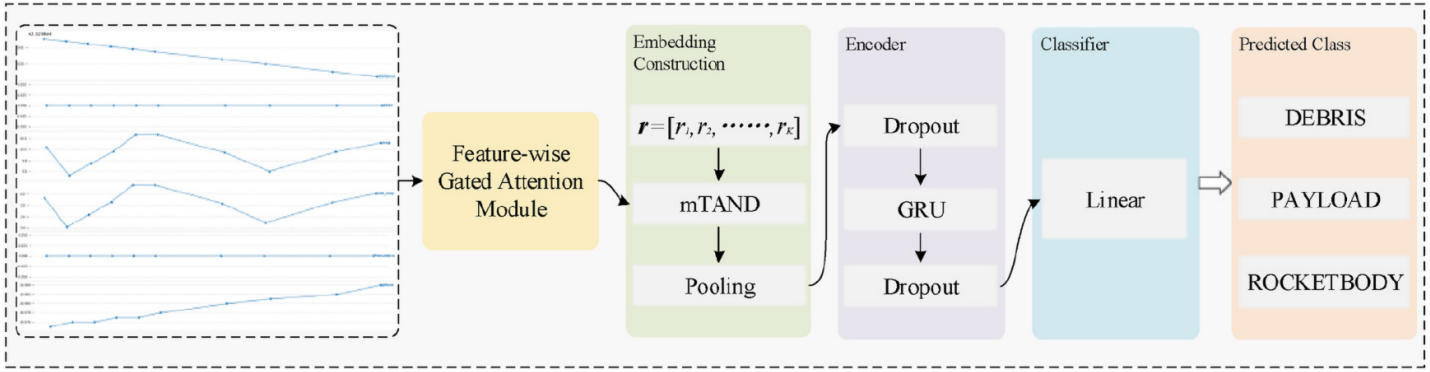


Fig. 10. Architecture of the improved mTAND network for RSO light curve classification.

Simulation and Analysis

Table 5
Validation accuracy and computational efficiency of the three models.

Network	Validation accuracy	#Parameters (M)	Average Inference Time(ms)	Throughput (samples/s)
TPCNN	0.8405	5.33	0.1647 ± 0.0056	6071.09
mTAND	0.8676	1.62	0.0492 ± 0.0092	20314.81
improved mTAND	**0.8811**	**0.146**	**0.0331 ± 0.0031**	**30253.48**

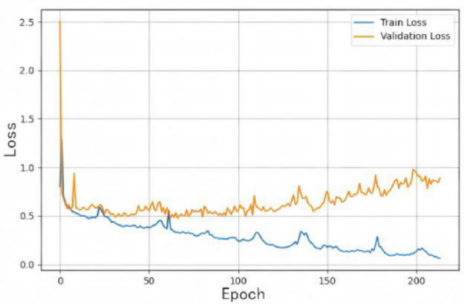
Table 6
Test results of the three networks.

Network	Mean accuracy	Mean precision	Mean recall	Mean F1-Score
TPCNN	0.7918 ± 0.02275	0.7931 ± 0.02282	0.7918 ± 0.02275	0.7910 ± 0.02332
mTAND	0.7807 ± 0.01677	0.7800 ± 0.01660	0.7807 ± 0.01677	0.7791 ± 0.01636
optimized mTAND	**0.8117 ± 0.00786**	**0.8131 ± 0.00789**	**0.8117 ± 0.00786**	**0.8110 ± 0.00807**

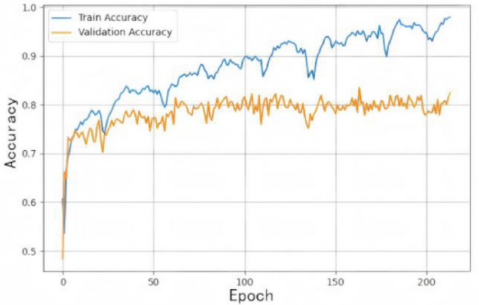
Table 7
Test performance of classical baseline models.

Model	Mean accuracy	Mean precision	Mean recall	Mean F1-Score
RNN	0.7829	0.8021	0.7829	0.7875
LSTM	0.7873	0.7990	0.7873	0.7897
GRU	**0.7961**	**0.8345**	**0.7961**	**0.8035**
1D-CNN	0.7873	0.8255	0.7873	0.7958

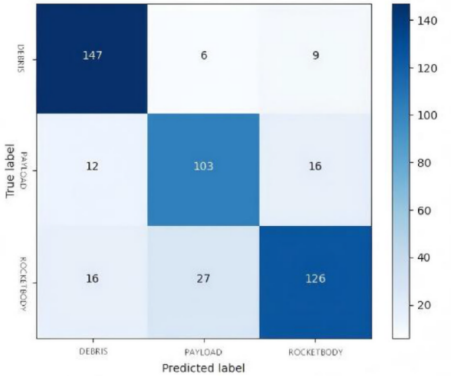
TPCNN Results



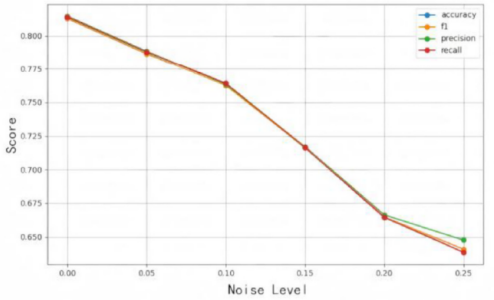
A. Training and validation loss of TPCNN model



B. Training and validation accuracy of TPCNN model



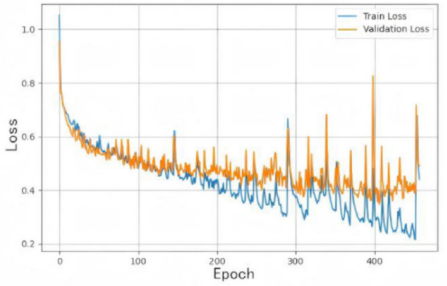
C. Test confusion matrix of the best TPCNN model



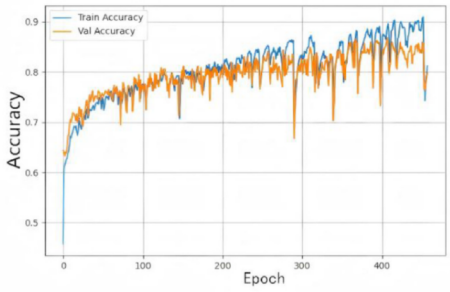
D. Robustness of the TPCNN model under different noise levels

Fig. 11. Performance of the TPCNN

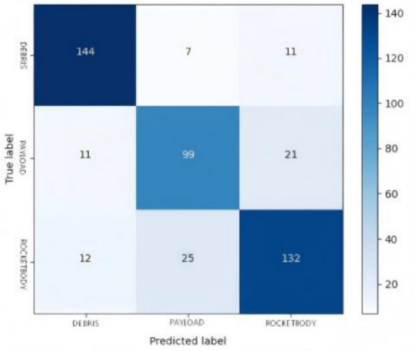
mTAND Results



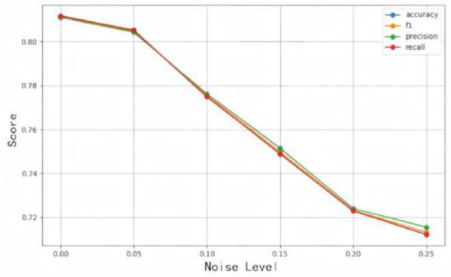
A. Training and validation loss of mTAND model



B. Training and validation accuracy of mTAND model



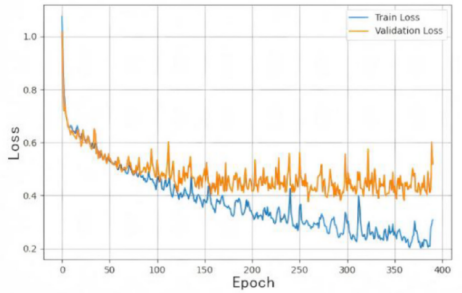
C. Test confusion matrix of the best mTAND model



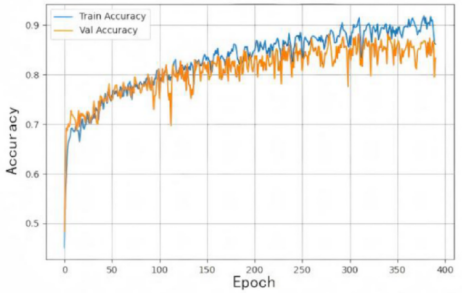
D. Robustness of the mTAND model under different noise levels

Fig. 12. Performance of the mTAND network.

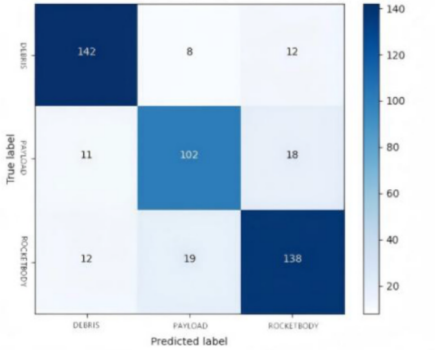
Improved mTAND Results



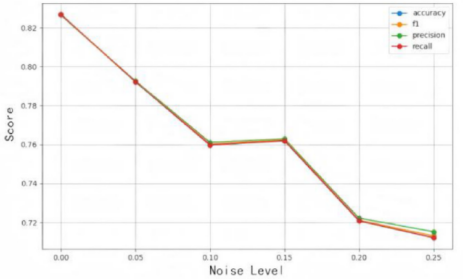
A. Training and validation loss of the improved mTAND model



B. Training and validation accuracy of the improved mTAND model



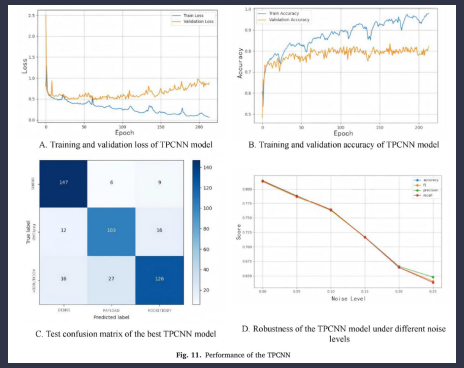
C. Test confusion matrix of improved mTAND model

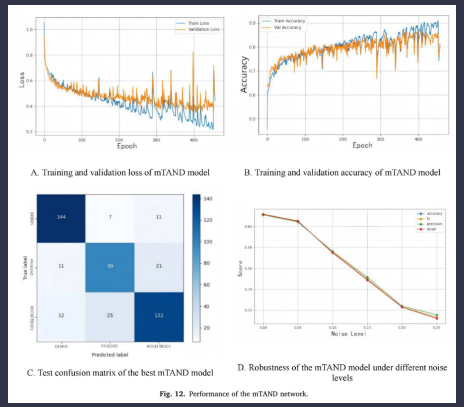


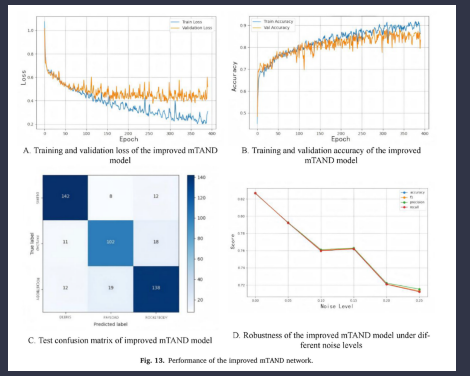
D. Robustness of the improved mTAND model under different noise levels

Fig. 13. Performance of the improved mTAND network.

Simulation and Analysis Combined







Attention mechanism not only outperforms standard neural networks but is cheaper and quicker to train, is smaller, and does not end up overfitting in longer training

Questions?



Sources

- **[1]** H. Zhao, X. Liu, G. Wang, and S. Li, “Deep learning-based classification of resident space objects with non-periodic light curves,” Engineering Applications of Artificial Intelligence, vol. 181, p. 115677, Oct. 2026, doi: 10.1016/j.engappai.2026.115677.
- **[2]** European Space Agency, “Space Environment Statistics,” sdup.esoc.esa.int, Mar. 31, 2025. https://sdup.esoc.esa.int/discosweb/statistics/ (accessed Oct. 04, 2026).
- **[3]** H. N. Russell, "On the light variations of asteroids and satellites," Astrophys. J., vol. 24, no. 1, pp. 1-18, Jul. 1906.
- **[4]** G. Valyavin, G. Beskin, A. Valeev, and A. M. Sobolev, "General view of the assembled entire Mini-MegaTORTORA system," in Exoplanet and Transient Event Investigation Project—Optical Facilities and Solutions, Dec. 2022. [Online]. Available: ResearchGate, researchgate.net.